

Workbook: Neuronal Integration

Week 7 workbook · BIO 004 Human Anatomy · Dr. Sharilyn Rennie · Solano Community College

A comprehensive workbook on how neurons integrate signals: the resting membrane, graded potentials, action potentials, and the basic anatomy and organization of neural circuits.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

Membrane potential · 2 questions

Q01 · DOK 1 · Recall

What is the resting membrane potential of a neuron, and what maintains it?

Q02 · DOK 1 · Recall

What does the sodium-potassium pump do?

Graded vs action potentials · 3 questions

Q03 · DOK 1 · Recall

What is a graded potential, where does it occur, and what does it do?

Q04 • DOK 1 • Recall

Describe the basic shape and meaning of an action potential.

Q05 • DOK 2 • Compare

Compare temporal and spatial summation of graded potentials.

Circuit organization • 3 questions**Q06 • DOK 1 • Recall**

Name the components of a basic reflex arc.

Q07 • DOK 2 • Compare

Compare divergent and convergent neural circuits.

Q08 • DOK 1 • Recall

What is a reverberating circuit, and where would it be useful?
